KAIZEN MATHS

Lesson starter

Poisson Distribution

Approximation and Model Choice

Name

S1

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 150 and p = 0.01. Use a Poisson approximation to estimate P(X = 0).

S2

Approximate the binomial model using a Poisson cumulative probability.

A rare event has probability 0.02 on each of 200 independent trials. Approximate the probability of at most 2 events.

S3

Choose the distribution by checking the context and assumptions.

Choose a suitable distribution model: A teacher records the number of typing errors on a randomly chosen page of text.

S4

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 200 and p = 0.015. Use a Poisson approximation to estimate P(X = 0).

KAIZEN MATHS

Lesson starter

Poisson Distribution

Approximation and Model Choice

Name

S1

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 150 and p = 0.01. Use a Poisson approximation to estimate P(X = 0).

S2

Approximate the binomial model using a Poisson cumulative probability.

A rare event has probability 0.02 on each of 200 independent trials. Approximate the probability of at most 2 events.

S3

Choose the distribution by checking the context and assumptions.

Choose a suitable distribution model: A teacher records the number of typing errors on a randomly chosen page of text.

S4

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 200 and p = 0.015. Use a Poisson approximation to estimate P(X = 0).

STA-3014-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Approximation and Model Choice

Name _____

EA1

Approximate the binomial model using a Poisson cumulative probability.

A rare event has probability 0.02 on each of 150 independent trials. Approximate the probability of at most 3 events.

EA2

Choose the distribution by checking the context and assumptions.

Choose a suitable distribution model: A call centre records the number of calls arriving in each 10-minute interval.

Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Approximation and Model Choice

Name _____

EB1

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 120 and p = 0.03. Use a Poisson approximation to estimate P(X = 4).

Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Approximation and Model Choice

Name _____

EA1

Approximate the binomial model using a Poisson cumulative probability.

A rare event has probability 0.02 on each of 150 independent trials. Approximate the probability of at most 3 events.

EA2

Choose the distribution by checking the context and assumptions.

Choose a suitable distribution model: A call centre records the number of calls arriving in each 10-minute interval.

STA-3014-L4 · EA

Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Approximation and Model Choice

Name _____

EB1

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 120 and p = 0.03. Use a Poisson approximation to estimate P(X = 4).

STA-3014-L4 · EB

Show your working

Teacher answers & working

Poisson Distribution · Approximation and Model Choice

STA-3014-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 150 and p = 0.01. Use a Poisson approximation to estimate P(X = 0).

Answer:

P(X = 0) approximately 0.2231.

Working:

For large n and small p, use X approximately Po(lambda) with lambda = np.

lambda = 150 x 0.01 = 1.5.

So X is approximated by Po(1.5).

P(X = 0) approximately $\frac{e^{-1.5}1.5^0}{0!}$ = 0.2231.

S2

Approximate the binomial model using a Poisson cumulative probability.

A rare event has probability 0.02 on each of 200 independent trials. Approximate the probability of at most 2 events.

Answer:

P(X ≤ 2) approximately 0.2381.

Working:

Let X be the number of rare events. The exact model is Bin(200, 0.02).

Since n is large and p is small, use the Poisson approximation with lambda = np.

lambda = 200 x 0.02 = 4.

P(X ≤ 2) approximately P(X = 0) + P(X = 1) + P(X = 2).

P(X ≤ 2) approximately 0.0183 + 0.0733 + 0.1465 = 0.2381.

S3

Choose the distribution by checking the context and assumptions.

Choose a suitable distribution model: A teacher records the number of typing errors on a randomly chosen page of text.

Answer:

Poisson distribution.

Working:

Identify what is being counted.

The variable counts random events in a fixed region.

Therefore a suitable model is: Poisson distribution.

Teacher answers & working

Poisson Distribution · Approximation and Model Choice

STA-3014-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 200 and p = 0.015. Use a Poisson approximation to estimate P(X = 0).

Answer:

P(X = 0) approximately 0.0498.

Working:

For large n and small p, use X approximately Po(lambda) with lambda = np.

lambda = 200 x 0.015 = 3.

So X is approximated by Po(3).

P(X = 0) approximately $\frac{e^{-3}3^0}{0!}$ = 0.0498.

EA1

Approximate the binomial model using a Poisson cumulative probability.

A rare event has probability 0.02 on each of 150 independent trials. Approximate the probability of at most 3 events.

Answer:

P(X ≤ 3) approximately 0.6472.

Working:

Let X be the number of rare events. The exact model is Bin(150, 0.02).

Since n is large and p is small, use the Poisson approximation with lambda = np.

lambda = 150 x 0.02 = 3.

P(X ≤ 3) approximately P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3).

P(X ≤ 3) approximately 0.0498 + 0.1494 + 0.2240 + 0.2240 = 0.6472.

EA2

Choose the distribution by checking the context and assumptions.

Choose a suitable distribution model: A call centre records the number of calls arriving in each 10-minute interval.

Answer:

Poisson distribution.

Working:

Identify what is being counted.

The variable counts events occurring randomly over a fixed interval.

Therefore a suitable model is: Poisson distribution.

Teacher answers & working

Poisson Distribution · Approximation and Model Choice

STA-3014-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Use lambda = np, then approximate with a Poisson distribution.

A binomial variable has n = 120 and p = 0.03. Use a Poisson approximation to estimate P(X = 4).

Answer:

P(X = 4) approximately 0.1912.

Working:

For large n and small p, use X approximately Po(lambda) with lambda = np.

lambda = 120 x 0.03 = 3.6.

So X is approximated by Po(3.6).

P(X = 4) approximately $\frac{e^{-3.6}3.6^4}{4!}$ = 0.1912.